

NOTES 20: CLT-BASED INFERENCE FOR MEANS

Stat 120 | Spring 2026

Prof Amanda Luby

CLT: The Central Limit Theorem (CLT) tells us that if the sample size is big enough and the sample is random,

$$\bar{X} \sim N(\text{---}, \text{---})$$

Example: Finding z^* with a table. What could you say about the value of the p-value for a z-score of 2.917?

percentage	percentile (qnorm(percentage))
90%	1.3
95%	1.6
97.5%	2.0
99%	2.3
99.5%	2.6



1 How big is big enough?

Rule of Thumb for Means:

2 How do we find the SE?

Standard Error for Means:

Idea: As n gets bigger, the SE gets _____. If _____ is small, the SE is also _____.

Example: Florida Lakes

$H_0 :$

$H_A :$

$\bar{X} =$

$s =$

Test stat:

p-value:

Example: Guinness Beer Acidity

$H_0 :$

$H_A :$

$\bar{X} =$

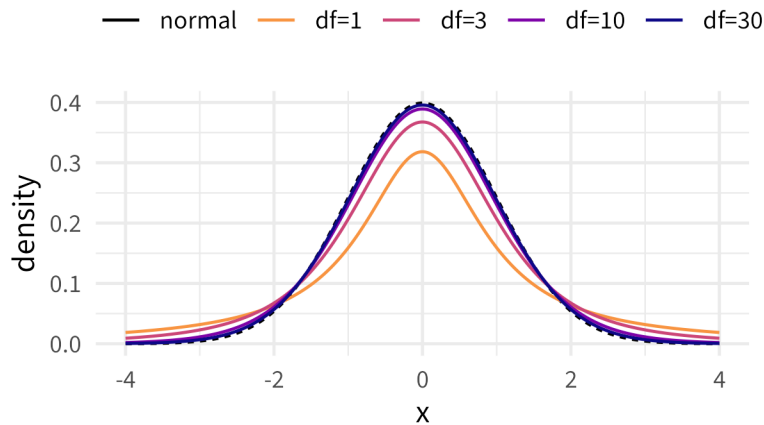
$s =$

Test stat:

p-value:

p-value (t-distribution):

3 The t-distribution



- When we divide by _____ instead of $\frac{\sigma}{\sqrt{n}}$, the test stat has a t-distribution instead of a $N(0,1)$
- The t-distribution depends on the “degrees of freedom” (_____)
- When df is _____, t-distribution has “heavier tails” than $N(0,1)$
- When df is _____, the t-distribution is approximately equal to $N(0,1)$

Example: Florida Lakes (again)

```
t = -1.75  
pt(t, df = 52)
```

```
[1] 0.04301
```

4 Summary

- Test stat for means: _____
- SE: _____
- Can “safely” use the CLT if _____
- If _____, we can still use the CLT if there are no outliers or extreme skew
- t-distribution is better to use, but for large sample sizes it will be close to the normal distribution
- Percentage of t-distribution below t -score: `pt(t-score, df = n-1)`
- Percentile t^* for a specific percentage: `qt(percentage, df = n-1)`

5 Group Problems

1. (Adapted from Exercise 6.128)

Plastic microparticles contaminate shorelines. Much of the pollution comes from washing fleece clothing. In a recent study, washing a fleece garment discharged on average $\bar{X} = 290$ fibers per liter of wastewater. The standard deviation was $s = 87.6$ fibers and the sample size was $n = 120$.

- (a) What is the estimated *standard error* of the average number of fibers discharged per liter of wastewater when washing a fleece garment?
- (b) The table below gives some percentiles of the t_{119} distribution. Use this information to construct a 99% confidence interval for the population mean. Interpret the interval in context.

percentage	percentile (qnorm(percentage))
90%	1.3
95%	1.6
97.5%	2.0
99%	2.3
99.5%	2.6

- (c) What sample size would we need if we wanted this interval to be *no wider* than 20?